

Problem J. Joker or Judger

Time Limit: 1 seconds | Memory Limit: 512 MB

Standard Input / Standard Output

As the chief judge of The 2026 ICPC Asia EC Regionals Online Contest (II), Mr. J has diligently finished preparing and reviewing all the problems. Today, the contest is finally about to begin!

The 2025 ICPC Asia East Continent Online Contest (II)

☆   [-828]

Since last year's contest was harshly criticized, Mr. J cares greatly about this year's reputation. Therefore, he has set himself a goal: this year's contest should not receive too many negative reviews. To quantify the negative impact of these reviews, Mr. J assigns a score to each type of negative review:

Type of negative review	Score
WrongProblem X	100
SameProblem X	30
UnreasonableProblemArrangement	10
UnreasonableLimitForProblem X	5
WeakTestsForProblem X	3
BadProblem X	1

If a negative review does not match any of the negative-review types in the table, its score is 0. Matching is case-sensitive, and the string must exactly match the corresponding form in the table. For a type containing X , X is a placeholder; when matching, it must be replaced by exactly one uppercase English letter from $\{A, B, \dots, L\}$.

If you need to copy the concrete string corresponding to a type of negative review, we recommend referring to the first test case in the sample.

Mr. J has an expected total negative-review score P in mind. If the total score of all negative reviews is greater than P , Mr. J is a Joker; otherwise, Mr. J remains a qualified Judger. You have now collected all n negative reviews. Determine whether Mr. J is a Joker or a Judger.

Input

The first line contains a positive integer T ($1 \leq T \leq 10^3$), denoting the number of test cases.

For each test case, the first line contains two integers n, P ($0 \leq n \leq 10^3, 0 \leq P \leq 10^5$), denoting the number of negative reviews and the expected total negative-review score, respectively.

The next n lines each contain a string S ($1 \leq |S| \leq 50$) consisting only of uppercase and lowercase English letters, denoting one negative review.

It is guaranteed that, within a single test file, the sum of n over all test cases does not exceed 10^3 .

Output

For each test case, if the total negative-review score is greater than P , output the string **Joker** on one line; otherwise, output the string **Judger** on one line.

Examples

Sample Input 1	Sample Output 1
5 6 0 WrongProblemA SameProblemA UnreasonableProblemArrangement UnreasonableLimitForProblemA WeakTestsForProblemA BadProblemA 6 105 WrongProblemA BadProblemC wrongProblemB WrongProblemN WrongProblemAA UnreasonableProblemArrangementExtra 7 49 SameProblemJ UnreasonableProblemArrangement UnreasonableLimitForProblemA WeakTestsForProblemC BadProblemH SameProblemj BadProblem 4 140 WrongProblemJ SameProblemA UnreasonableProblemArrangement BadProblemJ 2 1 BadProblemJ BadProblemJ	Joker Judger Judger Joker Joker

Note

For the first test case, the scores of the 6 negative reviews are 100, 30, 10, 5, 3, 1, respectively. Their total score is $149 > P$.

For the second test case, the scores of the 6 negative reviews are 100, 1, 0, 0, 0, 0, respectively. Their total score is $101 \leq P$.

For the third test case, the scores of the 7 negative reviews are 30, 10, 5, 3, 1, 0, 0, respectively. Their total score is $49 \leq P$.

For the fourth test case, the scores of the 4 negative reviews are 100, 30, 10, 1, respectively. Their total score is $141 > P$.

For the fifth test case, the scores of the 2 negative reviews are 1, 1, respectively. Their total score is $2 > P$. Note that duplicate negative reviews are also counted.